

Package ‘CSOA’

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Type Package

Title Calculate Per-Cell Gene Set Scores Using Cell Set Overlaps

Version 0.1.0

Description Cell Set Overlap Analysis (CSOA) is a tool for calculating per-cell gene signature scores in a scRNA-seq dataset. For each gene in the signature, CSOA constructs a set consisting of cells highly expressing the gene, after which all overlaps of pairs of cell sets are computed, ranked, filtered and scored. The CSOA per-cell score is calculated on the basis of the top cell set overlaps by summing up, over all overlaps, the product of the overlap score and the min-max-normalized expression of each of the two genes.

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Imports bayesbio,
dplyr,
ggeasy,
ggforce,
ggnewscale,
ggplot2,
ggraph,
ggrepel,
graphics,
grDevices,
kerntools,
methods,
qs,
reshape2,
Seurat,
SeuratObject,
SingleCellExperiment,
SummarizedExperiment,
sgof,
spatstat.utils,
stats,
stringr,
textshape,
tidygraph,
viridis,

wesanderson

Encoding UTF-8

RoxygenNote 7.3.2

Collate 'compute_cell_scores_tools.R'
 'convex_hull.R'
 'extra_filter.R'
 'gene_degrees.R'
 'gene_modules.R'
 'generate_overlaps_tools.R'
 'generics.R'
 'main.R'
 'multiple_gene_sets.R'
 'percentile_sets.R'
 'process_overlaps_tools.R'
 'utils.R'
 'visualization_aux.R'
 'visualization.R'

Suggests BiocStyle,
 knitr,
 patchwork,
 rmarkdown,
 scRNAseq,
 scuttle

biocViews Software,
 SingleCell

VignetteBuilder knitr

URL <https://github.com/andrei-stoica26/CS0A>

BugReports <https://github.com/andrei-stoica26/CS0A/issues>

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basicHeatmap	<i>Plot a simple heatmap</i>
--------------	------------------------------

Description

This function plots a simple heatmap, with clustering but no dendograms.

Usage

```
basicHeatmap(
  mat,
  aesNames = c("x", "y", "fill"),
  title = "Heatmap",
  axisTextSize = 7,
  palType = "fillCont",
  ...
)
```

Arguments

mat	A matrix.
aesNames	A character vector of size 3 representing the y, x and fill aes elements.
title	Plot title.
axisTextSize	Axis text size.
palType	Palette type.
...	Additional arguments passed to <code>wesBinaryGradient</code> .

Value

A ggplot object

breakWeakTies	<i>Remove overlap pairs with low Jaccard scores</i>
---------------	---

Description

This function iteratively removes all overlap pairs with Jaccard score below a fixed cutoff until no overlap pairs can be removed

Usage

```
breakWeakTies(overlapDF, cutoff = 1/3, doConnComp = FALSE)
```

Arguments

overlapDF	An overlap data frame
cutoff	A cutoff used in the filtering of edges with low Jaccard scores
doConnComp	Whether to calculate the connected components

Value

An overlap data frame in which edges with low Jaccard scores have been removed

byCorrectDF	<i>Adjust a dataframe column of p-values with Benjamini-Yekutieli</i>
-------------	---

Description

This function performs the Benjamini-Yekutieli correction for multiple testing in a dataframe column of p-values and filters the data-frame based on p-values.

Usage

```
byCorrectDF(df, pvalThr = 0.05, colStr = "pval")
```

Arguments

df	A dataframe with a column of p-values
pvalThr	p-value threshold
colStr	Name of the column of p-values

Value

The data frame with Benjamini-Yekutieli-corrected p-values

cellDistribution	<i>Show the distribution of cell sets among cells</i>
------------------	---

Description

This function returns a matrix that shows the presence of cell sets among cells.

Usage

```
cellDistribution(cellSets, allCells)
```

Arguments

cellSets	A list of character vectors.
allCells	Names of all cells in the dataset.

Value

A logical matrix with genes as rows and cells as columns.

cellSetsOverlaps	<i>Calculates the significance of overlaps of pairs of cells sets</i>
------------------	---

Description

This function computes the statistical significance of overlaps of pairs of cell sets.

Usage

```
cellSetsOverlaps(cellSets, nCells, pairs = NULL, overlapFileName = NULL)
```

Arguments

cellSets	A list of character arrays.
nCells	The total number of cells in the Seurat object.
pairs	Pairs of cell sets to be assessed. If NULL (as default), all pairs will be assessed.
overlapFileName	The name of the file where the overlap data frame will be saved. Default is NULL (the overlap data frame will not be saved).

Value

A data frame listing statistics for all cell set overlaps.

computeCellScores	<i>Assign a per-cell gene set score to Seurat object</i>
-------------------	--

Description

This function uses the scored data frame of overlaps to compute a CSOA score for each cell in a Seurat object

Usage

```
computeCellScores(
  overlapDF,
  normExp,
  cellNames,
  colStr = "CSOA",
  pairFileName = NULL,
  keepOverlapOrder = FALSE
)
```

Arguments

overlapDF	An overlap data frame.
normExp	A min-max normalized expression matrix of the genes involved in top overlaps.
cellNames	Cell names
colStr	The name of the column where CSOA results will be stored
pairFileName	The name of the file where the pair data frame will be saved. Default is NULL (the pair data frame will not be saved)
keepOverlapOrder	Keep the rank-based order of overlaps in the pair score file, as opposed to changing it to a pair score-based order. Ignored if pairFileName is NULL

Value

A Seurat object with a CSOA score assigned for each cell

computePairScores	<i>Compute aggregate gene pair scores</i>
-------------------	---

Description

This function assesses the relative contribution of each gene pair to the CSOA score

Usage

```
computePairScores(
  overlapDF,
  pcPairScores,
  pairFileName = NULL,
  keepOverlapOrder = FALSE
)
```

Arguments

overlapDF	An overlap data frame.
pcPairScores	A data frame of pair scores in each cell for each pair in the overlap data frame
pairFileName	The name of the file where the pair data frame will be saved. Default is NULL (the pair data frame will not be saved)
keepOverlapOrder	Keep the rank-based order of overlaps in the pair score file, as opposed to changing it to a pair score-based order. Ignored if pairFileName is NULL

Value

A data frame with overlap and pair scores and ranks

computePCPairScores	<i>Compute per-cell gene pair scores</i>
---------------------	--

Description

This function scores each gene pair corresponding to a top overlap in each cell.

Usage

```
computePCPairScores(overlapDF, normExp)
```

Arguments

overlapDF	An overlap data frame.
normExp	A min-max normalized expression matrix of the genes involved in top overlaps.

Value

A data frame with the same dimensions as normExp.

computePCSetScores	<i>Aggregate per-cell gene pair scores</i>
--------------------	--

Description

This function aggregates per-cell gene pair scores into per-cell gene signature scores

Usage

```
computePCSetScores(pcPairScores, cellNames, colStr = "CSOA")
```

Arguments

pcPairScores	A data frame of pair scores in each cell for each pair in the overlap data frame
cellNames	Cell names
colStr	The name of the column where CSOA results will be stored

Value

A data frame with the per-cell gene signature score as a column

connectedComponents	<i>Find the connected components of the graph determined by the overlaps</i>
---------------------	--

Description

This function finds the connected components of the graph having the filtered overlaps as edges

Usage

```
connectedComponents(df, colName = "component")
```

Arguments

df	A data frame with gene1 and gene2 columns.
colName	Name of the connected components column to be added.

Value

An overlap data frame with a column indicated the number of the connected component.

devPlot	<i>Saves plot or list of plots</i>
---------	------------------------------------

Description

This function saves a plot or list of plots as a pdf. Can also take as input a function with arguments that returns a ggplot object.

Usage

```
devPlot(plotObject, ...)  
  
## Default S3 method:  
devPlot(plotObject, ...)  
  
## S3 method for class '`function`'  
devPlot(plotObject, ...)  
  
## S3 method for class 'ggplot'  
devPlot(plotObject, ...)  
  
## S3 method for class 'list'  
devPlot(plotObject, ...)
```

Arguments

plotObject	A function, ggplot object, or list of ggplot objects
...	Arguments passed to other methods

Value

No value. This function is called for its side effect (saving a plot or a list of plots)

expMat	<i>Extracts the data expression matrix from object</i>
--------	--

Description

This function extracts the data expression matrix from object as a non-sparse matrix

Usage

```
expMat(scObj, ...)

## Default S3 method:
expMat(scObj, genes = NULL, ...)

## S3 method for class 'Seurat'
expMat(scObj, ...)

## S3 method for class 'SingleCellExperiment'
expMat(scObj, ...)

## S3 method for class 'dgCMatrix'
expMat(scObj, ...)

## S3 method for class 'matrix'
expMat(scObj, ...)
```

Arguments

scObj	A Seurat object, SingleCellExperiment object, or expression matrix
...	Arguments passed to other methods
genes	Genes retained in the expression matrix. If NULL, all genes will be retained

Value

An expression matrix.

featureWes

Improved feature plot with a highly distinctive color scheme

Description

This function customizes the appearance of Seurat's FeaturePlot for improved distinctiveness and aesthetics.

Usage

```
featureWes(
  seuratObj,
  feature,
  title = feature,
  idClass = NULL,
  labelSize = 3,
  titleSize = 12,
  ...
)
```

Arguments

seuratObj	A SeuratObj.
feature	Seurat feature.
title	Plot title.
idClass	Column to be used for labelling. If NULL, no column-based labels will be generated.
labelSize	Size of labels. Ignored if idClass is NULL.
titleSize	Title size.
...	Other arguments passed to wesBinaryGradient.

Value

A ggplot object.

filterOverlaps	<i>Filter cell set overlaps</i>
----------------	---------------------------------

Description

This function filters cell set overlaps after the overlap data frame has been ranked

Usage

```
filterOverlaps(overlapDF, firstOutRawRank = NULL)
```

Arguments

overlapDF	A ranked overlap data frame
firstOutRawRank	The raw aggregate rank of the first overlap that will be excluded

Value

A filtered overlap data frame

geneRadialPlot	<i>Gene circle plot for an overlap data frames</i>
----------------	--

Description

This functions draws a gene-circle plot for an overlap data frame.

Usage

```
geneRadialPlot(
  overlapObj,
  groupLegendName = NULL,
  groupNames = NULL,
  cutoff = NULL,
  title = "Top overlap genes plot",
  extraCircles = 2
)
```

Arguments

overlapObj	An overlap data frame or list of overlap data frames
groupLegendName	The title of the group legend; if NULL, no groups will be distinguished.
groupNames	Names of groups. If provided, must be a vector of the same length as the list of overlap data frames
cutoff	Number of retained edges from each overlap data frame after refiltering. If NULL (as default), no refiltering will be performed
title	Plot title.
extraCircles	Number of circles drawn beyond those needed to include the points representing the genes. Default is 0

Value

A ggplot object.

generateOverlaps	<i>Generate overlaps of cell sets for input genes</i>
------------------	---

Description

This function builds a set of cells for each input gene by first selecting all the cells expressing the gene and then retaining only the cells expressing the gene at the input percentile. Subsequently, overlaps of pairs of the constructed cell sets are assessed for statistical significance.

Usage

```
generateOverlaps(
  geneSetExp,
  percentile = 90,
  pairs = NULL,
  overlapFileName = NULL
)
```

Arguments

geneSetExp	A gene expression non-sparse matrix with the rows restricted to the genes for which cell sets will be computed
percentile	A non-negative number below 100
pairs	Pairs of cell sets to be assessed. If NULL (as default), all pairs will be assessed.
overlapFileName	The name of the file where the overlap data frame will be saved. Default is NULL (the overlap data frame will not be saved).

Value

A data frame listing statistics for all cell set overlaps

networkPlot	<i>Plot the overlaps as a network</i>
-------------	---------------------------------------

Description

This function plots the graph of the overlap data frame, with genes as vertices and overlaps as edges.

Usage

```
networkPlot(
  overlapDF,
  title = "Top overlaps network plot",
  rankCol = "rank",
  edgeScale = 2,
  nodePointSize = 10,
  nodeTextSize = 2.3,
  ...
)
```

Arguments

overlapDF	Overlap data frame.
title	Plot title.
rankCol	Name of the rank column.
edgeScale	Scaling factor used in generating edge weights.
nodePointSize	Point size of graph nodes.
nodeTextSize	Text size of graph nodes.
...	Additional parameters passed to titlePlot.

Value

A network plot.

overlapCutoffPlot	<i>Plot the selection of overlaps</i>
-------------------	---------------------------------------

Description

This plots shows the process of selecting the overlap rank cutoff, showcasing the convex hull of the points representing the frequencies of each rank.

Usage

```
overlapCutoffPlot(freqDF, rankCutoff, title = "Overlap cutoff plot")
```

Arguments

freqDF	A frequency data frame of overlap ranks.
rankCutoff	Rank cutoff
title	Plot title

Value

A ggplot object

overlapGenes	<i>Get all genes from an overlap data frame</i>
--------------	---

Description

This function gets all genes from an overlap data frame

Usage

```
overlapGenes(overlapDF)
```

Arguments

overlapDF A data frame created with generate_overlaps

Value

A character vector of genes

overlapPairs	<i>Extract gene pairs from overlap matrix</i>
--------------	---

Description

This function extracts the gene pairs from an overlap matrix

Usage

```
overlapPairs(overlapDF)
```

Arguments

overlapDF A data frame created with generate_overlaps

Value

A list of gene pairs

percentileSets	<i>Generates cell expressing input genes at an input percentile</i>
----------------	---

Description

This function first finds, for each input gene, the cells with a non-zero expression of the gene. Subsequently, each cell set is filtered as to contain only the cells showcasing the input percentile of the expression of the gene defining the set.

Usage

```
percentileSets(geneSetExp, percentile = 90)
```

Arguments

geneSetExp	A gene expression non-sparse matrix with the rows restricted to the genes for which cell sets will be computed
percentile	A non-negative number below 100

Value

A named list of character vectors of length equaling the number of input genes, storing, for each gene, the cells showing the input percentile in terms of their expression of the gene

prepareFiltering	<i>Find the raw aggregate rank of the highest non-top overlap</i>
------------------	---

Description

This function finds the raw aggregate rank of the highest non-top overlap.

Usage

```
prepareFiltering(overlapDF, saveCutoffPlot = FALSE)
```

Arguments

overlapDF	A ranked overlap data frame
saveCutoffPlot	Whether to save overlap cutoff plot

Value

A numeric value

processOverlaps	<i>Process data frame of overlaps of cell sets</i>
-----------------	--

Description

This function filters, ranks and scores previously generated overlaps of cell sets

Usage

```
processOverlaps(
  overlapDF,
  pvalThr = 0.05,
  saveCutoffPlot = FALSE,
  jaccardCutoff = NULL,
  osMethod = "log"
)
```

Arguments

overlapDF	A data frame created with generate_overlaps
pvalThr	p-value threshold
saveCutoffPlot	Whether to save overlap cutoff plot
jaccardCutoff	A cutoff used in the filtering of edges with low Jaccard scores. NULL by default (no filtering of such edges will be performed).
osMethod	The method used to compute overlap scores. Options are 'log' and 'minmax'.

Value

A data frame consisting of filtered, ranked and scored cell sets overlaps

qGrab	<i>Read and delete a .qs file</i>
-------	-----------------------------------

Description

This functions reads a .qs file, deletes it, and returns its content

Usage

```
qGrab(qsFile)
```

Arguments

qsFile	Name of .qs file including its path
--------	-------------------------------------

Value

The content of the .qs file

rankOverlaps	<i>Rank cell set overlaps</i>
--------------	-------------------------------

Description

This function ranks cell set overlaps by taking the average of the adjusted p-value rank and the ratio of shared cells over expected shared cells rank.

Usage

```
rankOverlaps(overlapDF)
```

Arguments

overlapDF A data frame created with generate_overlaps

Value

A data frame with ranked overlaps

runCSOA	<i>Run the CSOA pipeline</i>
---------	------------------------------

Description

This function generates cell set overlaps for an input gene set based on percentiles of gene expression, computes the significance of these overlaps, ranks, filters and scores the overlaps based on this significance, and builds a per-cell score by summing the products of the scores of these overlaps and the custom-normalized per-cell expressions of the corresponding pairs of genes.

Usage

```
runCSOA(
  scObj,
  genes,
  colStr = "CSOA",
  percentile = 90,
  overlapFileName = NULL,
  pvalThr = 0.05,
  saveCutoffPlot = FALSE,
  jaccardCutoff = NULL,
  osMethod = "log",
  pairFileName = NULL,
  keepOverlapOrder = FALSE
)
```

Arguments

scObj	A Seurat object, SingleCellExperiment object, or expression matrix
genes	Vector of genes. Must include at least two genes
colStr	The name of the column where CSOA results will be stored
percentile	A non-negative number below 100
overlapFileName	The name of the file where the overlap data frame will be saved. Default is NULL (the overlap data frame will not be saved).
pvalThr	p-value threshold
saveCutoffPlot	Whether to save overlap cutoff plot
jaccardCutoff	A cutoff used in the filtering of edges with low Jaccard scores. NULL by default (no filtering of such edges will be performed).
osMethod	The method used to compute overlap scores. Options are 'log' and 'minmax'.
pairFileName	The name of the file where the pair data frame will be saved. Default is NULL (the pair data frame will not be saved)
keepOverlapOrder	Keep the rank-based order of overlaps in the pair score file, as opposed to changing it to a pair score-based order. Ignored if pairFileName is NULL

Value

An object of the same class as scObj with a CSOA score assigned for each cell

runCSOAMultiple	<i>Run the CSOA pipeline for multiple gene sets</i>
-----------------	---

Description

This function generates cell set overlaps for input gene sets based on percentiles of gene expression, computes the significance of these overlaps, ranks, filters and scores the overlaps based on this significance, and builds a per-cell score by summing the products of the scores of these overlaps and the custom-normalized per-cell expressions of the corresponding pairs of genes.

Usage

```
runCSOAMultiple(
  scObj,
  geneSets,
  geneSetNames,
  percentile = 90,
  overlapFileName = NULL,
  pvalThr = 0.05,
  saveCutoffPlot = FALSE,
  jaccardCutoff = NULL,
```

```

    osMethod = "log",
    pairFileTemplate = NULL,
    keepOverlapOrder = FALSE
  )

```

Arguments

scObj	A Seurat object, SingleCellExperiment object, or expression matrix
geneSets	List of character vectors
geneSetNames	Character vector of names of gene sets
percentile	A non-negative number below 100
overlapFileName	The name of the file where the overlap data frame will be saved. Default is NULL (the overlap data frame will not be saved).
pvalThr	p-value threshold
saveCutoffPlot	Whether to save overlap cutoff plot
jaccardCutoff	A cutoff used in the filtering of edges with low Jaccard scores. NULL by default (no filtering of such edges will be performed).
osMethod	The method used to compute overlap scores. Options are 'log' and 'minmax'.
pairFileTemplate	Character object used in the naming of the files where the pair data frames will be saved. Default is NULL (the pair data frames will not be saved)
keepOverlapOrder	Keep the rank-based order of overlaps in the pair score file, as opposed to changing it to a pair score-based order. Ignored if pairFileName is NULL

Value

An object of the same class as scObj with per-gene-set CSOA scores assigned for each cell

scoreCells	<i>Generate CSOA scores from overlap data frame</i>
------------	---

Description

This function computes per-cell CSOA scores after the overlap data frame has been generated.

Usage

```

scoreCells(
  geneSetExp,
  overlapDF,
  colStr = "CSOA",
  pvalThr = 0.05,
  saveCutoffPlot = FALSE,

```

```

    jaccardCutoff = NULL,
    osMethod = "log",
    pairFileName = NULL,
    keepOverlapOrder = FALSE
  )

```

Arguments

geneSetExp	A gene expression non-sparse matrix with the rows restricted to the genes for which cell sets will be computed
overlapDF	A data frame created with generate_overlaps
colStr	The name of the column where CSOA results will be stored
pvalThr	p-value threshold
saveCutoffPlot	Whether to save overlap cutoff plot
jaccardCutoff	A cutoff used in the filtering of edges with low Jaccard scores. NULL by default (no filtering of such edges will be performed).
osMethod	The method used to compute overlap scores. Options are 'log' and 'minmax'.
pairFileName	The name of the file where the pair data frame will be saved. Default is NULL (the pair data frame will not be saved)
keepOverlapOrder	Keep the rank-based order of overlaps in the pair score file, as opposed to changing it to a pair score-based order. Ignored if pairFileName is NULL

Value

A data frame with a column corresponding to the CSOA scores

scoreCellsMultiple	<i>Generate CSOA scores from overlap data frame for multiple gene sets</i>
--------------------	--

Description

This function scores an overlap data frame generated using multiple gene sets. The overlap data frame is split based on the overlaps corresponding to each gene set and scored, and the output is rejoined as a data frame.

Usage

```

scoreCellsMultiple(
  geneSetExp,
  overlapDF,
  setPairs,
  geneSetNames,
  pvalThr = 0.05,
  saveCutoffPlot = FALSE,

```

```

    jaccardCutoff = NULL,
    osMethod = "log",
    pairFileTemplate = NULL,
    keepOverlapOrder = FALSE
  )

```

Arguments

geneSetExp	A gene expression non-sparse matrix with the rows restricted to the genes for which cell sets will be computed
overlapDF	A data frame created with generate_overlaps
setPairs	A list of overlaps corresponding to each input gene set
geneSetNames	Character vector of names of gene sets
pvalThr	p-value threshold
saveCutoffPlot	Whether to save overlap cutoff plot
jaccardCutoff	A cutoff used in the filtering of edges with low Jaccard scores. NULL by default (no filtering of such edges will be performed).
osMethod	The method used to compute overlap scores. Options are 'log' and 'minmax'.
pairFileTemplate	Character object used in the naming of the files where the pair data frames will be saved. Default is NULL (the pair data frames will not be saved)
keepOverlapOrder	Keep the rank-based order of overlaps in the pair score file, as opposed to changing it to a pair score-based order. Ignored if pairFileTemplate is NULL

Value

A data frame whose columns correspond to the CSOA scores of the input gene sets

scoreModules	<i>Run CSOA separately on the connected components of the overlap graph</i>
--------------	---

Description

This function runs CSOA on the connected components of the graph having the filtered overlaps as edges

Usage

```
scoreModules(scObj, df, components, colStrTemplate = "CSOA_component", ...)
```

Arguments

scObj	A Seurat object, SingleCellExperiment object, or expression matrix
df	A data frame with gene1, gene2 and component columns
components	Vector of connected components that will be scored
colStrTemplate	Character used in the naming of the component gene sets
...	Additional parameters to other functions

Value

An object of the same class as scObj with CSOA scores corresponding to the genes defining each connected components assigned for each cell

scoreOverlaps	<i>Score cell set overlaps</i>
---------------	--------------------------------

Description

This function scores cell set overlaps based on their ranks. The score of the top overlap is set to 1, and the score decreases logarithmically towards 0, which corresponds to the score of the first overlap not included in the filtered data frame (i.e., if the filtered data frame contains 100 overlaps, the 101st overlap corresponds to a score of 0).

Usage

```
scoreOverlaps(overlapDF, osMethod = "log", firstOutRawRank = NULL)
```

Arguments

overlapDF	A filtered overlap data frame.
osMethod	The method used to compute overlap scores. Options are 'log' and 'minmax'.
firstOutRawRank	The raw rank of the highest-ranked overlap that was not recorded among top overlaps. Ignored if osMethod is set to 'log'.

Value

A data frame with ranked overlaps.

storeCellScores	<i>Output CSOA scores</i>
-----------------	---------------------------

Description

This function returns the output generated by CSOA

Usage

```
storeCellScores(scObj, ...)

## Default S3 method:
storeCellScores(scObj, scoreDF, ...)

## S3 method for class 'Seurat'
storeCellScores(scObj, scoreDF, ...)

## S3 method for class 'SingleCellExperiment'
storeCellScores(scObj, scoreDF, altExpName = "CSOA", ...)

## S3 method for class 'matrix'
storeCellScores(scObj, scoreDF, ...)
```

Arguments

scObj	A Seurat object, SingleCellExperiment object, or expression matrix
...	Arguments passed to other methods
scoreDF	Dataframe of CSOA scores
altExpName	Name of the matrix storing CSOA scores

Value

A CSOA score assigned for each cell

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